

Algebraic and Geometric Reconstruction: A Bridge via Conditional Variance

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Abstract

Paper IV introduces two witnesses for finite-sample reconstruction: the algebraic witness $\delta(L)$ (worst-case σ -algebra approximation error) and the geometric witness $(\mu \otimes \mu)(R_L)$ (mass of unseparated pairs under the lag- L equivalence relation). This note identifies the exact object mediating between them — the *conditional variance of indicators* — and establishes a two-sided comparison theorem. The easy direction requires no conditions. The hard direction requires that large fibres of the delay map be internally mixed, a condition implied by ergodicity but strictly weaker. As a corollary, reconstruction is equivalent to divergence of the collision entropy of the delay-vector distribution, giving an information-theoretic characterisation of reconstruction not present in Papers I–IV.

1 Setup

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $h \in L^\infty(\mu)$. For $L \geq 0$ define the *observable σ -algebra at lag L* :

$$\mathcal{O}_h^{(L)} := \sigma(h \circ T^0, h \circ T^1, \dots, h \circ T^L)$$

and the *lag- L delay map* $\Phi_h^{(L)} : X \rightarrow \mathbb{R}^{L+1}$, $\Phi_h^{(L)}(x) = (h(x), h(Tx), \dots, h(T^L x))$.

Definition 1.1 (Algebraic witness). The *σ -algebra approximation error at lag L* is

$$\delta(L) := \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E).$$

Definition 1.2 (Equivalence relation and geometric witness). Define the lag- L equivalence relation $x \sim_L x'$ iff $\Phi_h^{(L)}(x) = \Phi_h^{(L)}(x')$, and the corresponding unseparated-pairs set:

$$R_L := \{(x, x') \in X \times X : x \sim_L x'\}.$$

The *geometric witness* is $(\mu \otimes \mu)(R_L)$.

Definition 1.3 (Fibre structure). For $z \in \mathbb{R}^{L+1}$, let $F_z := (\Phi_h^{(L)})^{-1}(z)$ be the fibre over z , $p_z := \mu(F_z)$ the fibre mass, and μ_z the conditional measure on F_z . The push-forward $\nu_L := (\Phi_h^{(L)})_* \mu$ is the distribution of delay vectors.

Three facts are immediate from the definitions:

- (a) $R_L = \bigsqcup_z F_z \times F_z$, so $(\mu \otimes \mu)(R_L) = \int p_z^2 d\nu_L(z) = \sum_z p_z^2$ (in the discrete-fibre case).
- (b) The conditional variance identity: for any $S \in \mathcal{B}$,

$$\mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})] = \int p_z \cdot \mu_z(S) \mu_z(S^c) d\nu_L(z).$$

- (c) $\delta(L) = \sup_S \mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})]$, since $\mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})] = \|\mathbf{1}_S - \mathbb{E}[\mathbf{1}_S \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2$ and $\delta(L) = \sup_S \|\mathbf{1}_S - \mathbb{E}[\mathbf{1}_S \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2$ (Paper IV, Lemma 3.1).

2 The Bridge Identity

Lemma 2.1 (Conditional variance identity). *For any $S \in \mathcal{B}$:*

$$\mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})] = \frac{1}{2} \int_{R_L} |\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 d(\mu \otimes \mu)(x, x').$$

Proof. By disintegration, $(\mu \otimes \mu)|_{R_L} = \int \mu_z \otimes \mu_z d\nu_L(z)$. On each fibre F_z :

$$\frac{1}{2} \int_{F_z \times F_z} |\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 d(\mu_z \otimes \mu_z) = \mu_z(S) \mu_z(S^c).$$

(Expand: $\int \int |\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 = 2[\mu_z(S)(1 - \mu_z(S))]$.) Integrating over fibres with weight $p_z = \mu(F_z)$:

$$\frac{1}{2} \int_{R_L} |\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 d(\mu \otimes \mu) = \int p_z \cdot \mu_z(S) \mu_z(S^c) d\nu_L(z) = \mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})]. \quad \square$$

Remark 2.2. Lemma 2.1 makes precise why the conditional variance is the right mediating object: its left-hand side is the quantity appearing in $\delta(L)$ (algebraic), and its right-hand side is an S -weighted version of $(\mu \otimes \mu)(R_L)$ (geometric). Both measure the same failure of separation, but with different aggregation: $\delta(L)$ takes the worst case over S ; $(\mu \otimes \mu)(R_L)$ counts unseparated pairs with equal weight.

3 The Two-Sided Comparison

Theorem 3.1 (Easy direction — unconditional).

$$\delta(L) \leq \frac{1}{2} (\mu \otimes \mu)(R_L).$$

Proof. From Lemma 2.1 and $|\mathbf{1}_S(x) - \mathbf{1}_S(x')|^2 \leq 1$:

$$\mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O}_h^{(L)})] \leq \frac{1}{2} (\mu \otimes \mu)(R_L)$$

for every S . Taking the supremum over S gives $\delta(L) \leq \frac{1}{2} (\mu \otimes \mu)(R_L)$. $\square$

Remark 3.2. This says: large unseparated-pair mass forces large worst-case error. No dynamical conditions are needed. The constant $\frac{1}{2}$ is sharp: take μ uniform on two points, $T = \text{id}$, h constant. Then $R_L = X \times X$, $(\mu \otimes \mu)(R_L) = 1$, and $\delta(L) = \frac{1}{2}$.

For the reverse direction, the identity in Lemma 2.1 shows that the gap between $(\mu \otimes \mu)(R_L) = \int p_z^2 d\nu_L(z)$ and $\delta(L) = \sup_S \int p_z \mu_z(S) \mu_z(S^c) d\nu_L(z)$ is controlled by whether large fibres (p_z large) are internally mixed.

Definition 3.3 (Fibre mixing condition). Say that $S^* \in \mathcal{B}$ is *c-mixing across fibres* if there exists $c > 0$ such that

$$\mu_z(S^*) \mu_z((S^*)^c) \geq c \cdot p_z \quad \text{for } \nu_L\text{-a.e. } z \text{ with } p_z > 0.$$

Theorem 3.4 (Hard direction — conditional). *If the event S^* achieving $\delta(L)$ is c-mixing across fibres, then*

$$(\mu \otimes \mu)(R_L) \leq \frac{1}{c} \delta(L).$$

Proof. From the fibre mixing condition:

$$\delta(L) \geq \int p_z \cdot \mu_z(S^*) \mu_z((S^*)^c) d\nu_L(z) \geq c \int p_z^2 d\nu_L(z) = c (\mu \otimes \mu)(R_L). \quad \square$$

Remark 3.5 (Fibre mixing vs ergodicity). The fibre mixing condition (Definition 3.3) is strictly weaker than ergodicity. It requires only that the adversarial event S^* intersects every positive-mass fibre in a non-trivial fraction. Ergodicity implies this: under an ergodic T , if S^* achieves $\delta(L) > 0$ then S^* is not $\mathcal{O}_h^{(L)}$ -measurable, so S^* must straddle some fibres, and by ergodicity the mass of non-straddled fibres is zero. But ergodicity is not necessary: the condition fails only when S^* is “aligned” with the fibre structure in the sense that large fibres are entirely inside or outside S^* .

Corollary 3.6 (Equivalence under fibre mixing). *Under the fibre mixing condition:*

$$c(\mu \otimes \mu)(R_L) \leq \delta(L) \leq \frac{1}{2}(\mu \otimes \mu)(R_L).$$

In particular:

$$\delta(L) \rightarrow 0 \iff (\mu \otimes \mu)(R_L) \rightarrow 0.$$

Proof. Theorems 3.1 and 3.4 combined. □

4 Entropy Characterisation of Reconstruction

The geometric witness $(\mu \otimes \mu)(R_L) = \int p_z^2 d\nu_L(z)$ is the collision probability of the delay-vector distribution ν_L . Its logarithm is the (negative) *Rényi-2 entropy* (collision entropy) of ν_L :

$$H_2(\nu_L) := -\log((\mu \otimes \mu)(R_L)) = -\log \int p_z^2 d\nu_L(z).$$

Corollary 4.1 (Entropy characterisation of reconstruction). *Under the fibre mixing condition of Definition 3.3:*

$$\delta(L) \rightarrow 0 \iff H_2(\nu_L) \rightarrow +\infty.$$

That is, reconstruction holds if and only if the collision entropy of the delay-vector distribution diverges.

Proof. $\delta(L) \rightarrow 0 \iff (\mu \otimes \mu)(R_L) \rightarrow 0$ (Corollary 3.6) $\iff e^{-H_2(\nu_L)} \rightarrow 0 \iff H_2(\nu_L) \rightarrow +\infty$. □

Remark 4.2 (Interpretation). Corollary 4.1 gives a purely information-theoretic characterisation of reconstruction: the observable h reconstructs the system $(X, \mathcal{B}, \mu, T)$ at lag L if and only if the distribution of the delay vector $\Phi_h^{(L)}(x)$ becomes increasingly spread — in the sense of Rényi-2 entropy — as L grows.

This characterisation is not present in Papers I–IV. It does not require:

- a metric on X ,
- smoothness of T or h ,
- a mixing rate,
- or the Takens embedding conditions.

The collision entropy $H_2(\nu_L)$ is estimable from data: it is the negative log of $\sum_z$ (fraction of sample in fibre z)², a standard diversity index computable from the delay-vector histogram. Under Paper IV’s standing hypotheses (compact smooth X , smooth positive μ , $T \in C^r$), the fibre mixing condition holds under ergodicity, and the entropy characterisation then provides a third, data-driven witness for reconstruction complementing $\delta(L)$ and the delay-map separation.

Remark 4.3 (Relation to Paper IV witnesses). The three witnesses now in scope are:

- (i) $\hat{\delta}(L, n)$ — the algebraic witness (Paper IV, §5);
- (ii) $\hat{d}_L(x, x')$ — the delay-map separation (Paper IV, §6);
- (iii) $\hat{H}_2(\nu_L^{(n)}) := -\log(\frac{1}{n^2} \sum_{i,j} \mathbf{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)])$ — the empirical collision entropy.

All three are computable from the time series alone. Corollaries 3.6 and 4.1 show that (i) and (iii) converge to zero (resp. diverge) together under fibre mixing, making them asymptotically equivalent witnesses for reconstruction. Witness (ii) provides the geometric certificate that the state space is being faithfully separated.